



Data Management

Nguyen Ngoc Thao
nnthao@fit.hcmus.edu.vn

Content outline

- Data in the world
- Data repository techniques
- NoSQL databases

Global Data Creation is About to Explode

Actual and forecast amount of data created worldwide 2010-2035 (in zettabytes)



@StatistaCharts

Source: Statista Digital Economy Compass 2019

statista

A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion - fuelled by internet of things and the use of connected devices - are hard to comprehend, particularly when looked at in the context of one day

500m

tweets are sent every day

Twitter



4PB

of data created by Facebook, including

350m photos

100m hours of video watch time

Facebook Research

294bn

billion emails are sent

Radiant Group

320bn

emails to be sent each day by 2021

306bn

emails to be sent each day by 2020

3.9bn

people use emails

4TB

of data produced by a connected car

Intel

DEMYSIFYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit	Value	Size
b bit	0 or 1	1/8 of a byte
B byte	8 bits	1 byte
KB kilobyte	1,000 bytes	1,000 bytes
MB megabyte	1,000 ² bytes	1,000,000 bytes
GB gigabyte	1,000 ³ bytes	1,000,000,000 bytes
TB terabyte	1,000 ⁴ bytes	1,000,000,000,000 bytes
PB petabyte	1,000 ⁵ bytes	1,000,000,000,000,000 bytes
EB exabyte	1,000 ⁶ bytes	1,000,000,000,000,000,000 bytes
ZB zettabyte	1,000 ⁷ bytes	1,000,000,000,000,000,000,000 bytes
YB yottabyte	1,000 ⁸ bytes	1,000,000,000,000,000,000,000,000 bytes

*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

65bn

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook



463EB

of data will be created every day by 2025

IEC

95m

photos and videos are shared on Instagram

Instagram Business



28PB

to be generated from wearable devices by 2020

Statista



Searches made a day

5bn

Searches made a day from Google

3.5bn

Smart Insights



ACCUMULATED DIGITAL UNIVERSE OF DATA

4.4ZB

44ZB

PwC

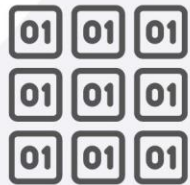
2013

2020

RACONTEUR

From **80 to 90 percent** of data generated and collected by organizations is **unstructured**.

Structured data



Characteristics

- Predefined data models
- Easy to search
- Text-based
- Shows what's happening

Resides in

- Relational databases
- Data warehouses

Stored in

- Rows and columns

Examples

- Dates, phone numbers, social security numbers, customer names, transaction info

Unstructured data



Characteristics

- No predefined data models
- Difficult to search
- Text, pdf, images, video
- Shows the why

Resides in

- Applications
- Data warehouses and lakes

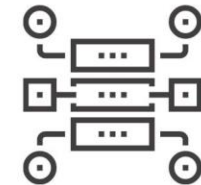
Stored in

- Various forms

Examples

- Documents, emails and messages, conversation transcripts, image files, open-ended survey answers

Semi-structured data



Characteristics

- Loosely organized
- Meta-level structure that can contain unstructured data
- HTML, XML, JSON

Resides in

- Relational databases
- Tagged-text format

Stored in

- Abstracts & figures

Examples

- Server logs, tweets organized by hashtags, emails sorting by folders (inbox; sent; draft)

- Data in the world has evolved with new characteristics.



Volume



Velocity



Variety



Veracity



Value

Image credit: Mailjet

- New technologies for handling data efficiently are needed.

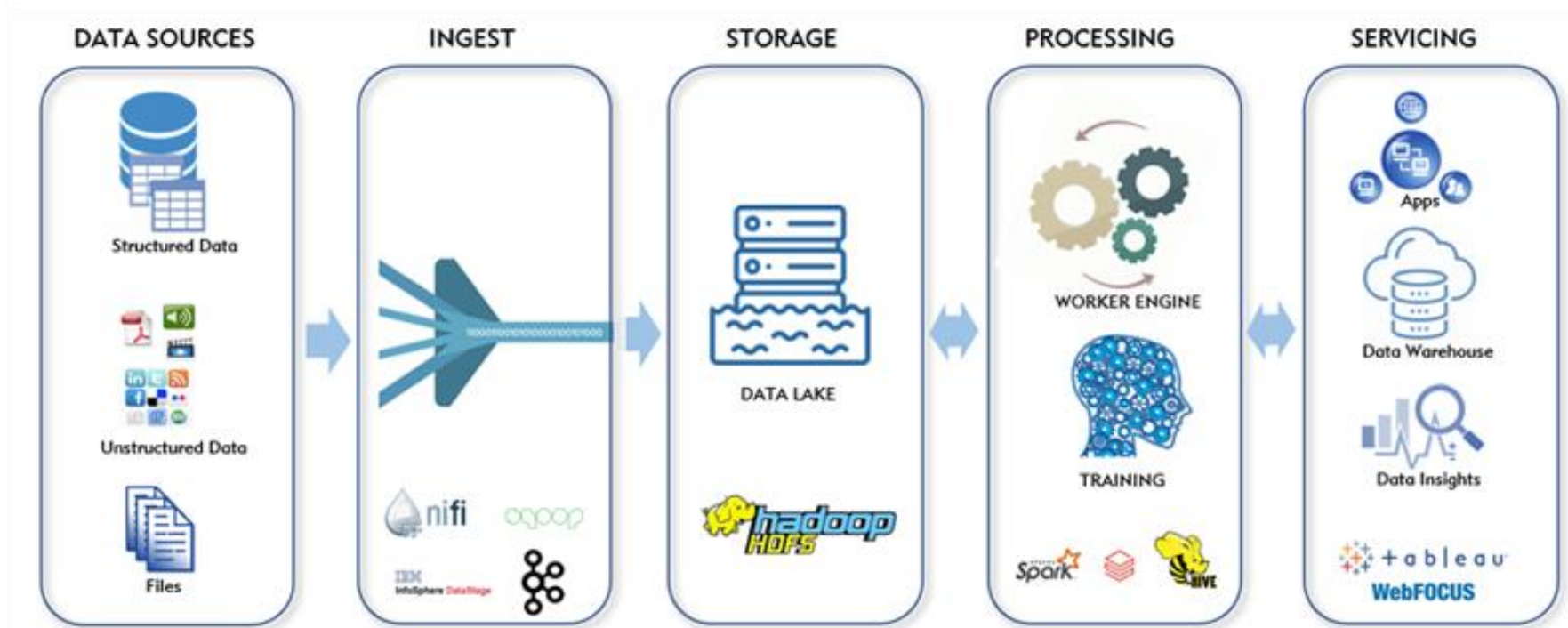
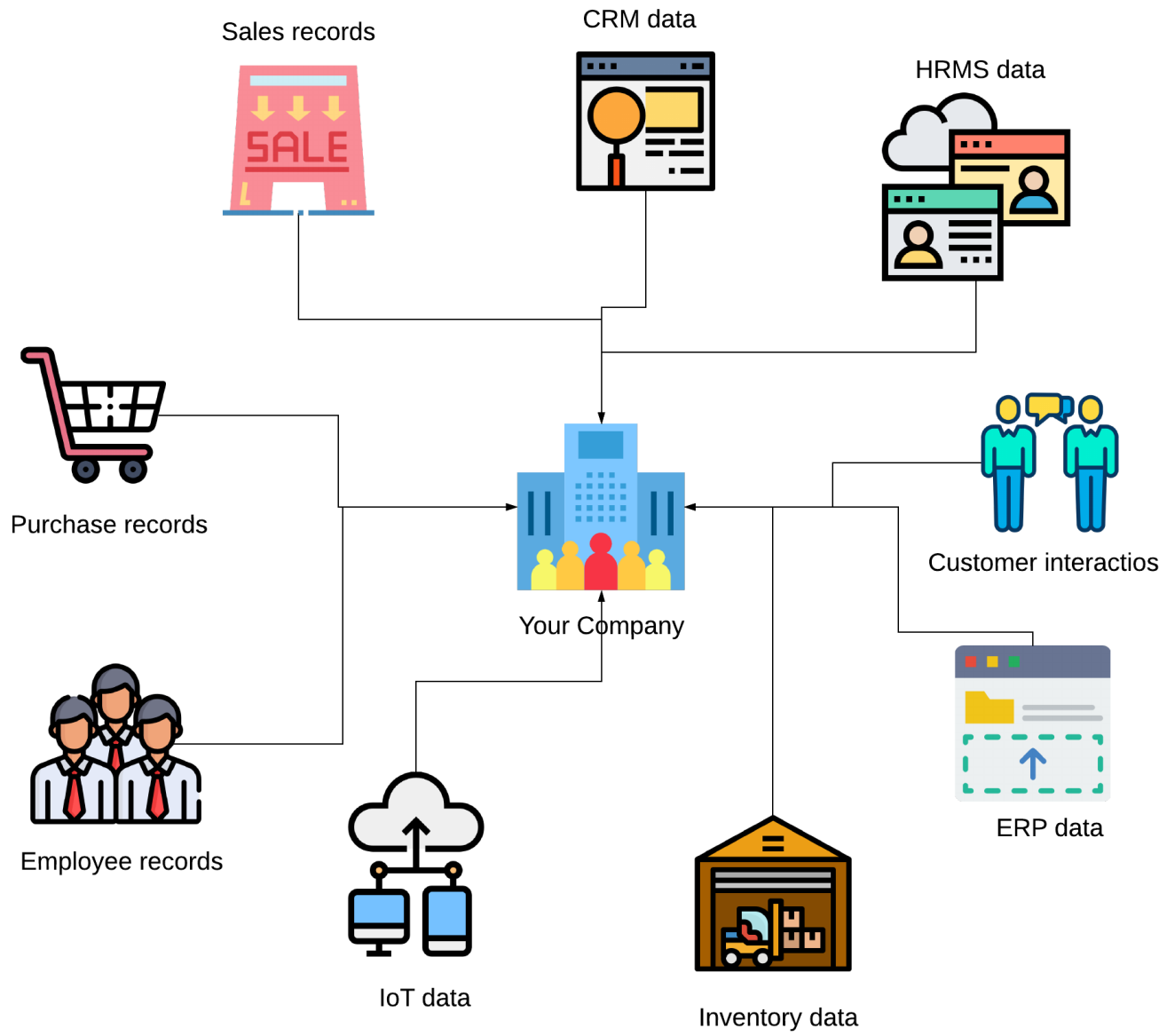


Image credit: Narwal Inc.

Data repository techniques

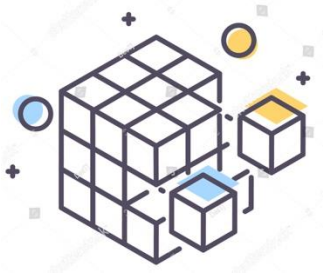


The data in an enterprise may come from different sources and in various formats.



Data warehouse

- A **data warehouse** is a data management system designed to **support analytics and decision makings**.
- It contains extensive historical data from various sources (e.g., log files and transaction applications).
- The key features of a data warehouse include



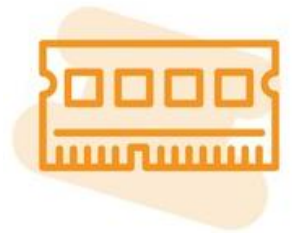
Subject-oriented



Integrated

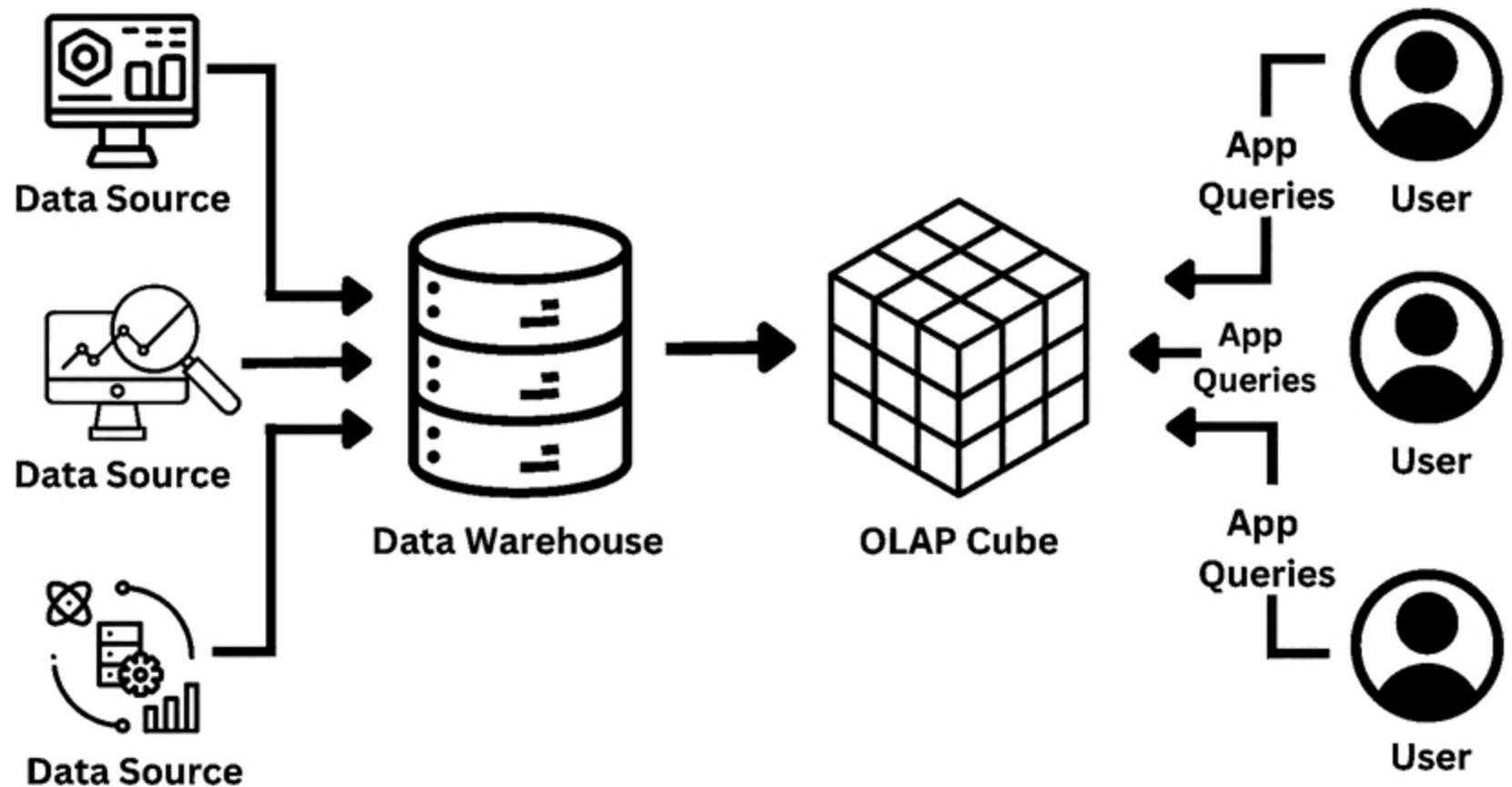


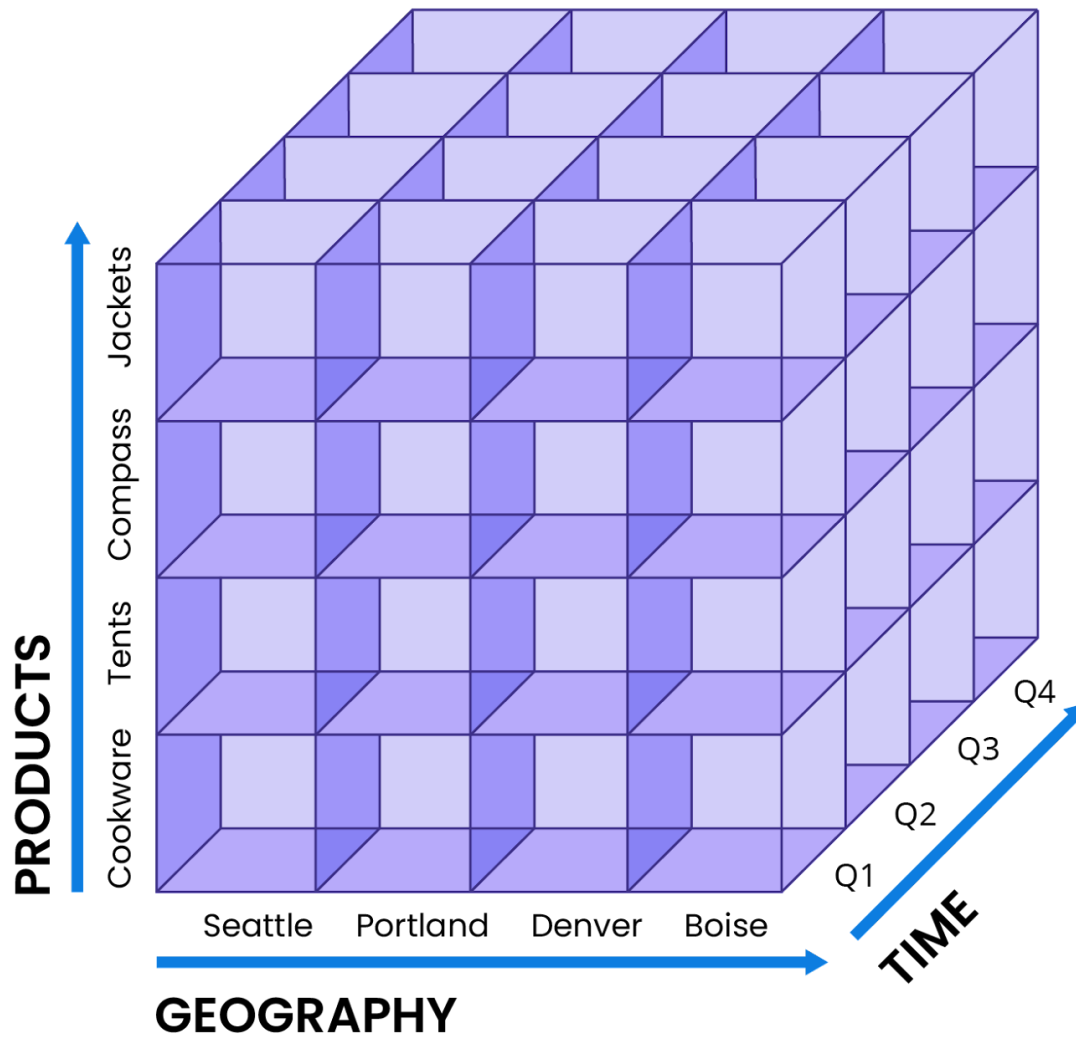
Time-variant



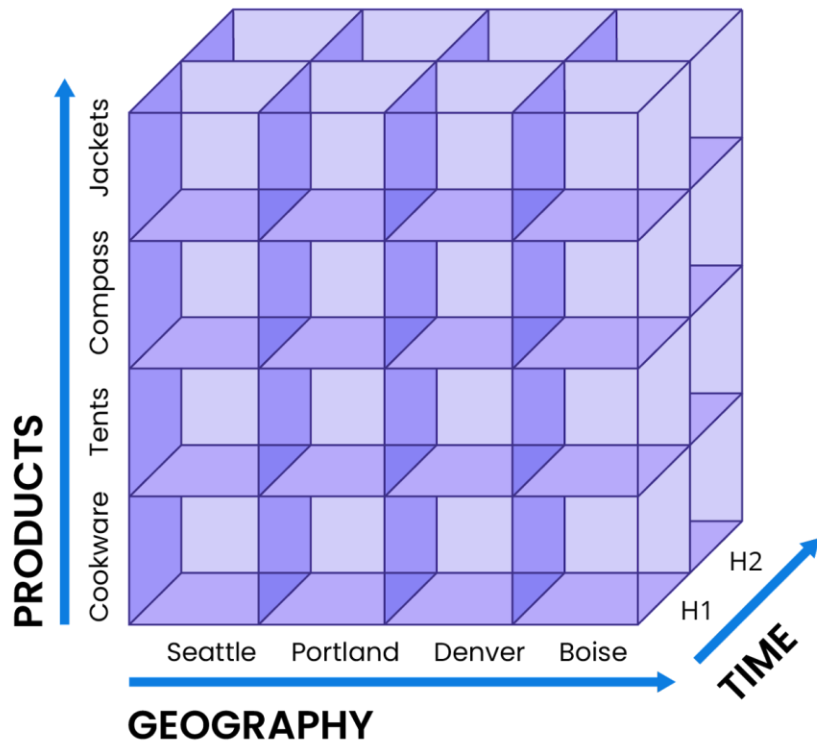
Nonvolatile

ONLINE ANALYTICAL PROCESSING

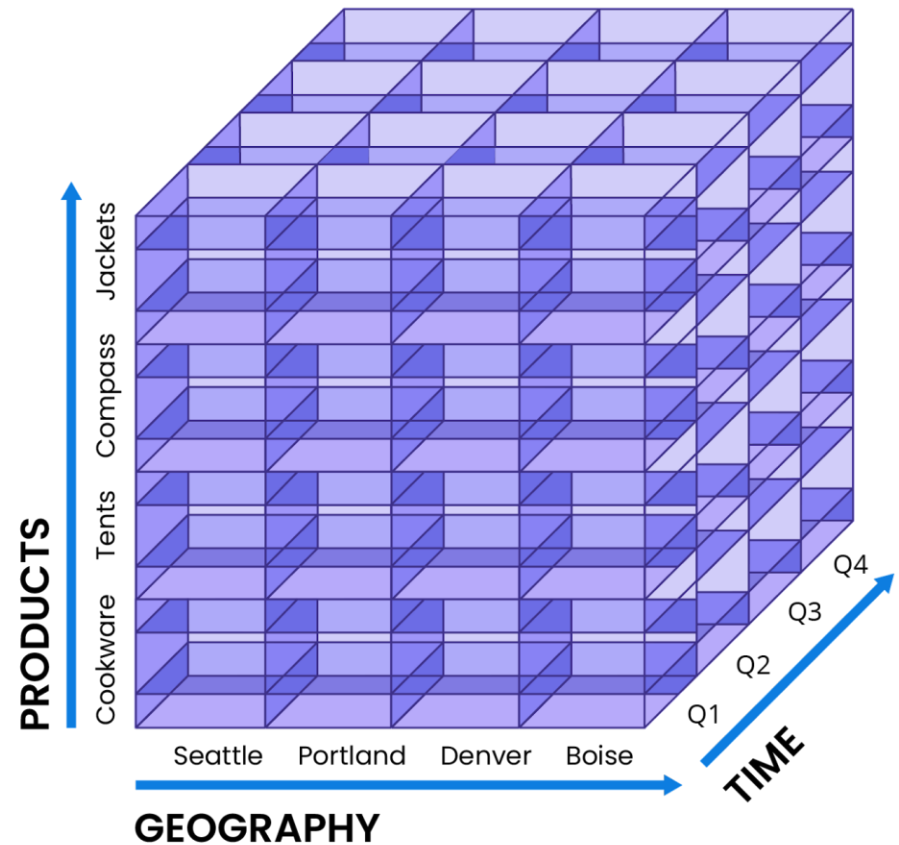




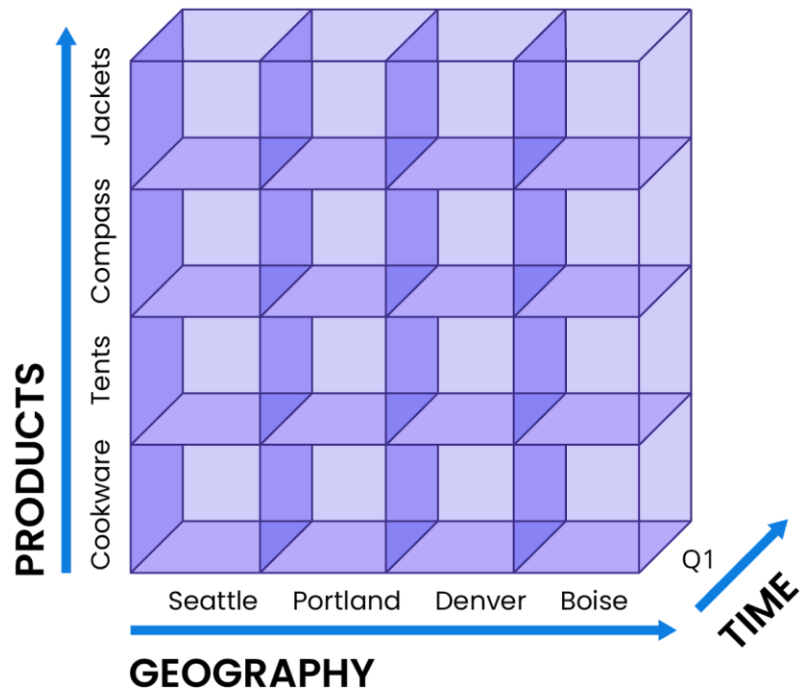
Original data cube



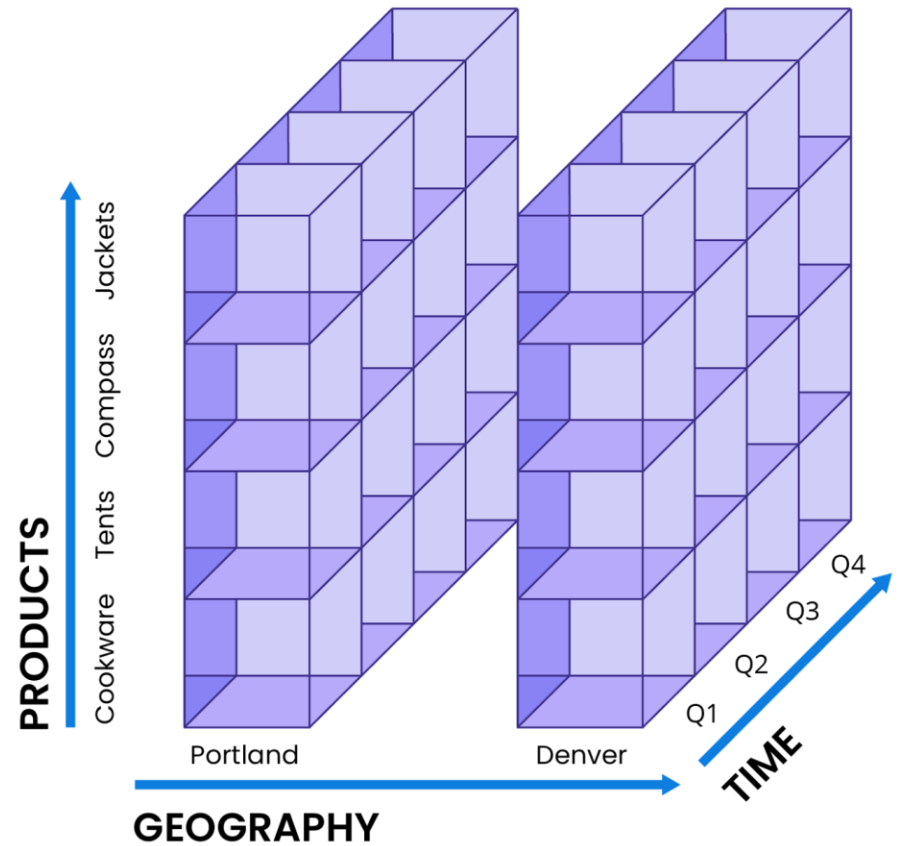
Roll up



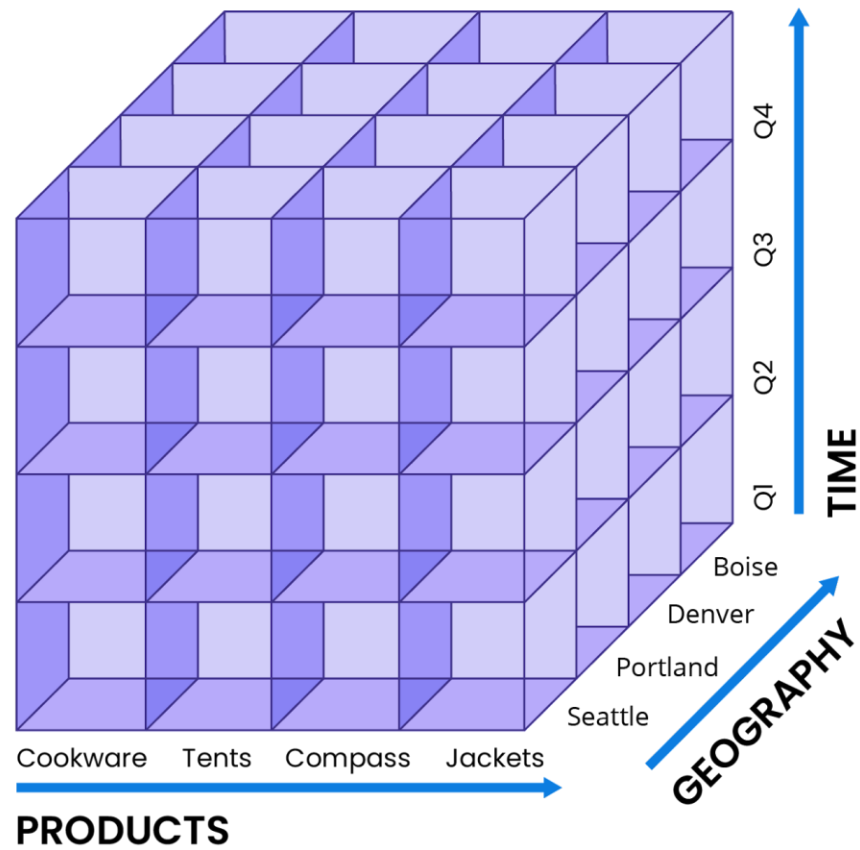
Drill down



Slice



Dice



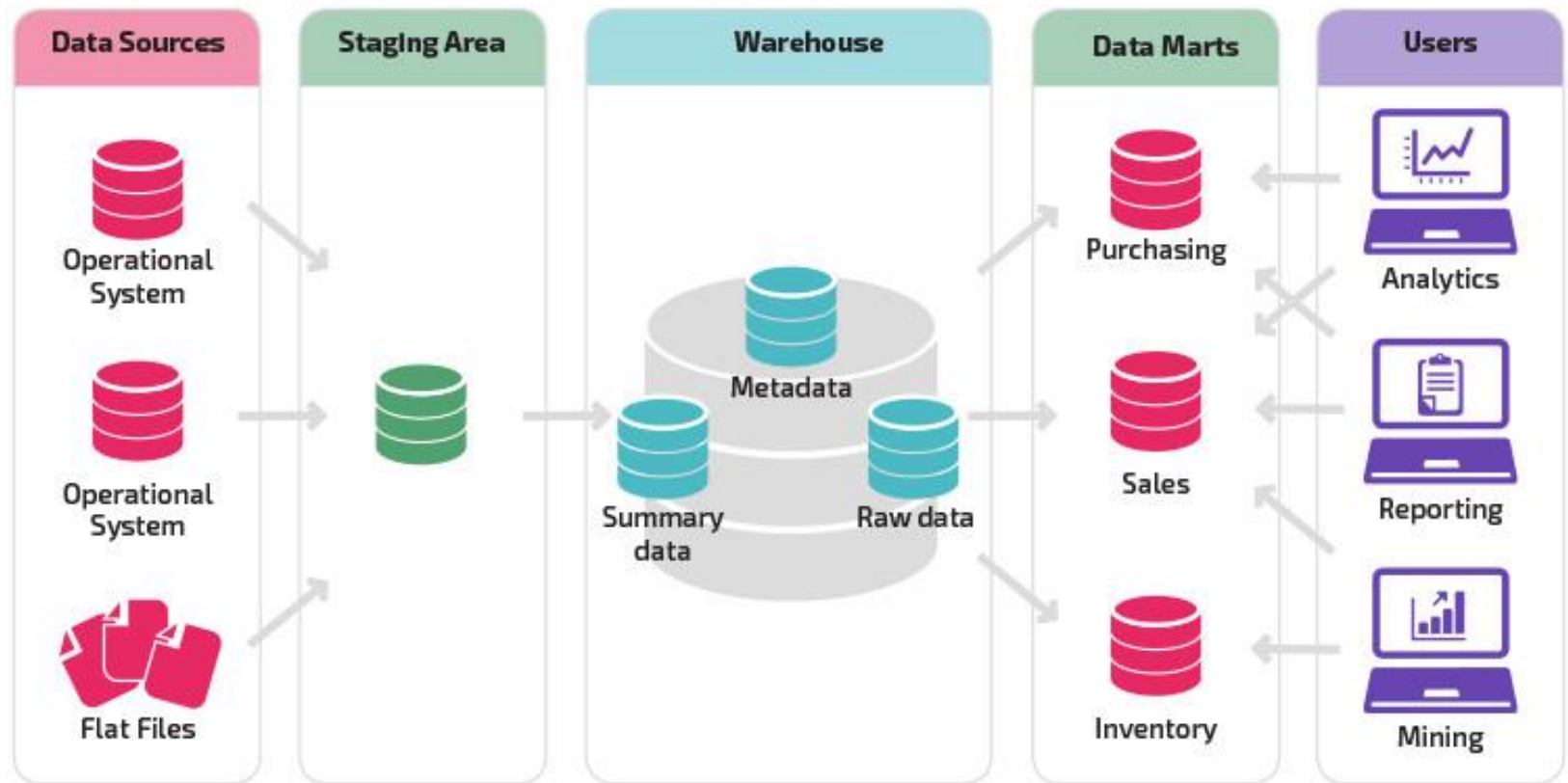
Pivot

OLAP vs. OLTP

Online Analytical Processing (OLAP)	Online Transactional Processing (OLTP)
In-depth data analysis for BI	Day-to-day operations of an organization
OLAP often works with historical data	OLTP usually handles real-time data
Execute complex operations, not as quickly and not on vast data	Read, write, and delete data rapidly on high-volume data
Involve large amounts of parallel users and queries	Not as many users and queries

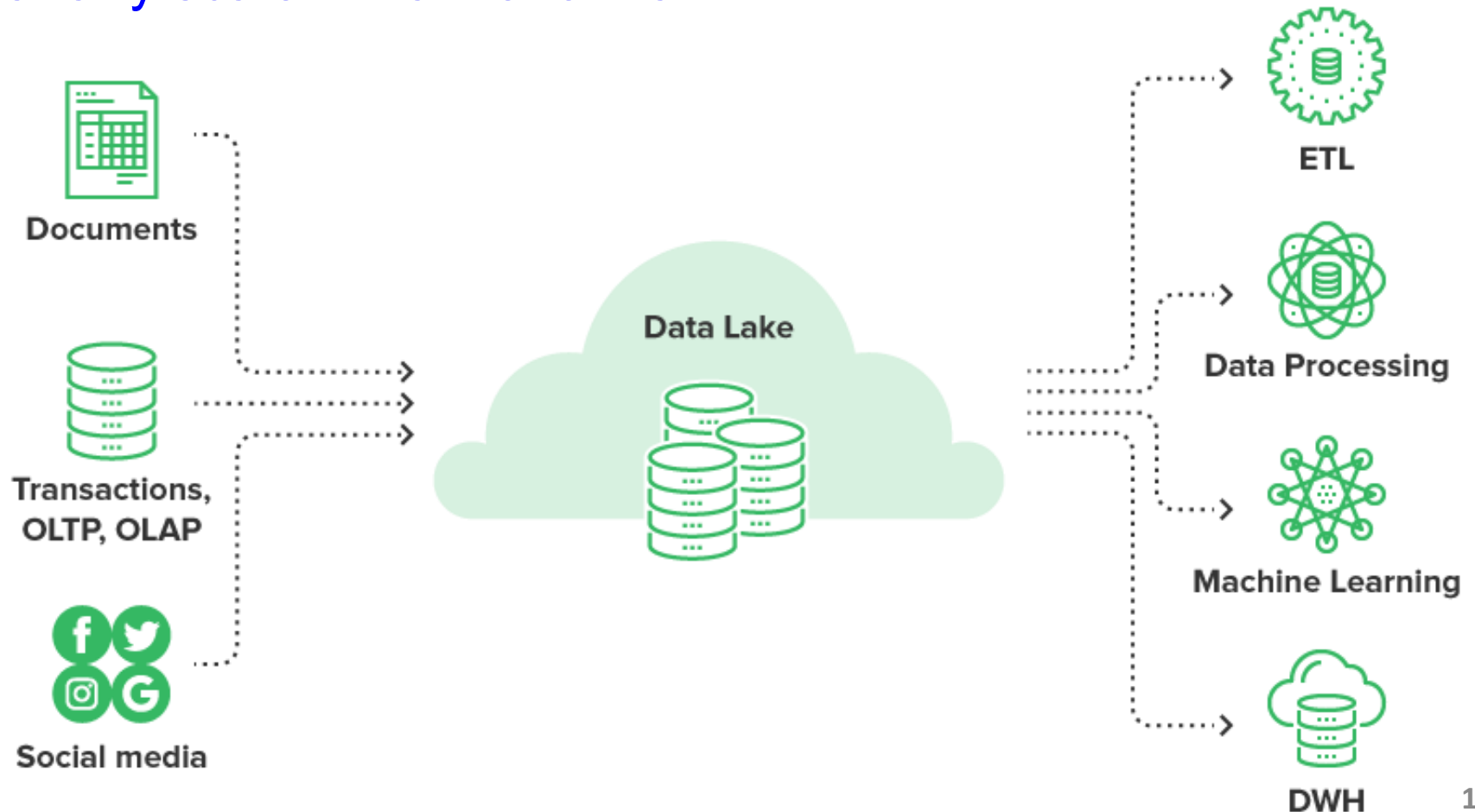
Data mart

- A **data mart** functions the same as a data warehouse does but within a much more limited scope.



Data lake

- A **data lake** is a centralized repository that stores **all the data** at any scale in native format.



Data lake vs. Data warehouse

Characteristics	Data Warehouse	Data Lake
Data	Relational from transactional systems, operational databases, and line of business applications	Non-relational and relational from IoT devices, web sites, mobile apps, social media, and corporate applications
Schema	Designed prior to the DW implementation (schema-on-write)	Written at the time of analysis (schema-on-read)
Price / Performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data quality	Highly curated data that serves as the central version of the truth	Any data that may or may not be curated (i.e., raw data)
Users	Business analysts	Data scientists, Data developers, and Business analysts (using curated data)
Analytics	Batch reporting, BI and visualizations	Machine Learning, Predictive analytics, data discovery and profiling

Source: aws.amazon.com

Data warehouse tools

teradata.

ORACLE
EXADATA

IBM Db2

 **amazon**
REDSHIFT

 **Google**
BigQuery

 **snowflake**

Data lake tools



AWS Lake Formation



Informatica™
INTELLIGENT DATA LAKE

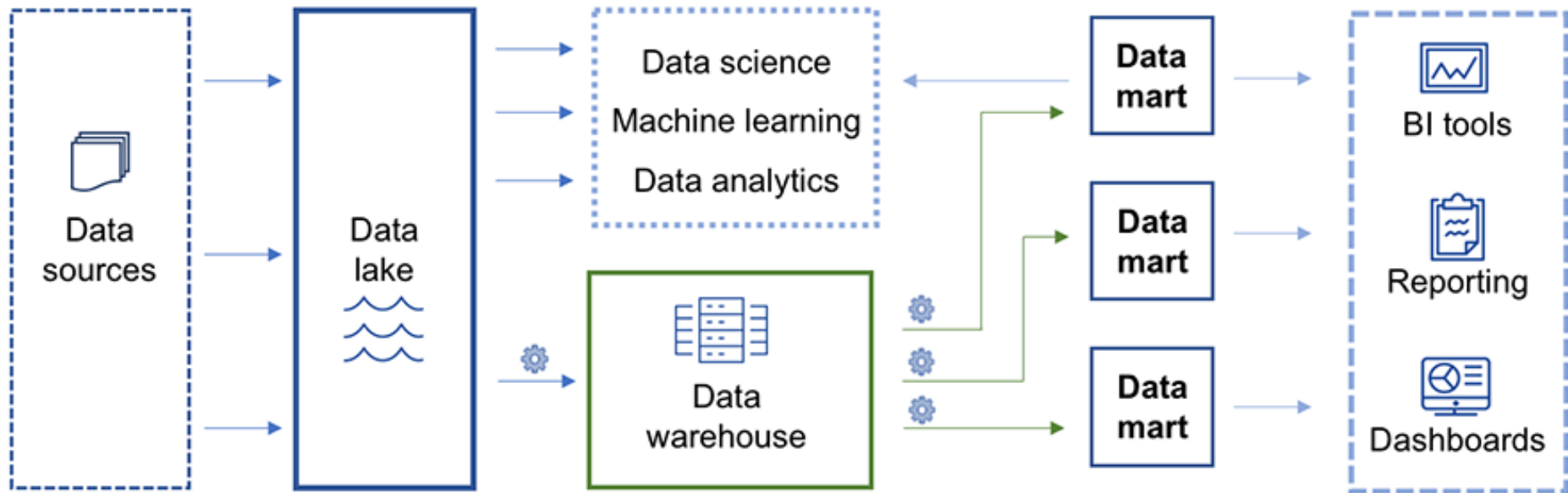


 **Qubole®**
THE OPEN DATA LAKE COMPANY

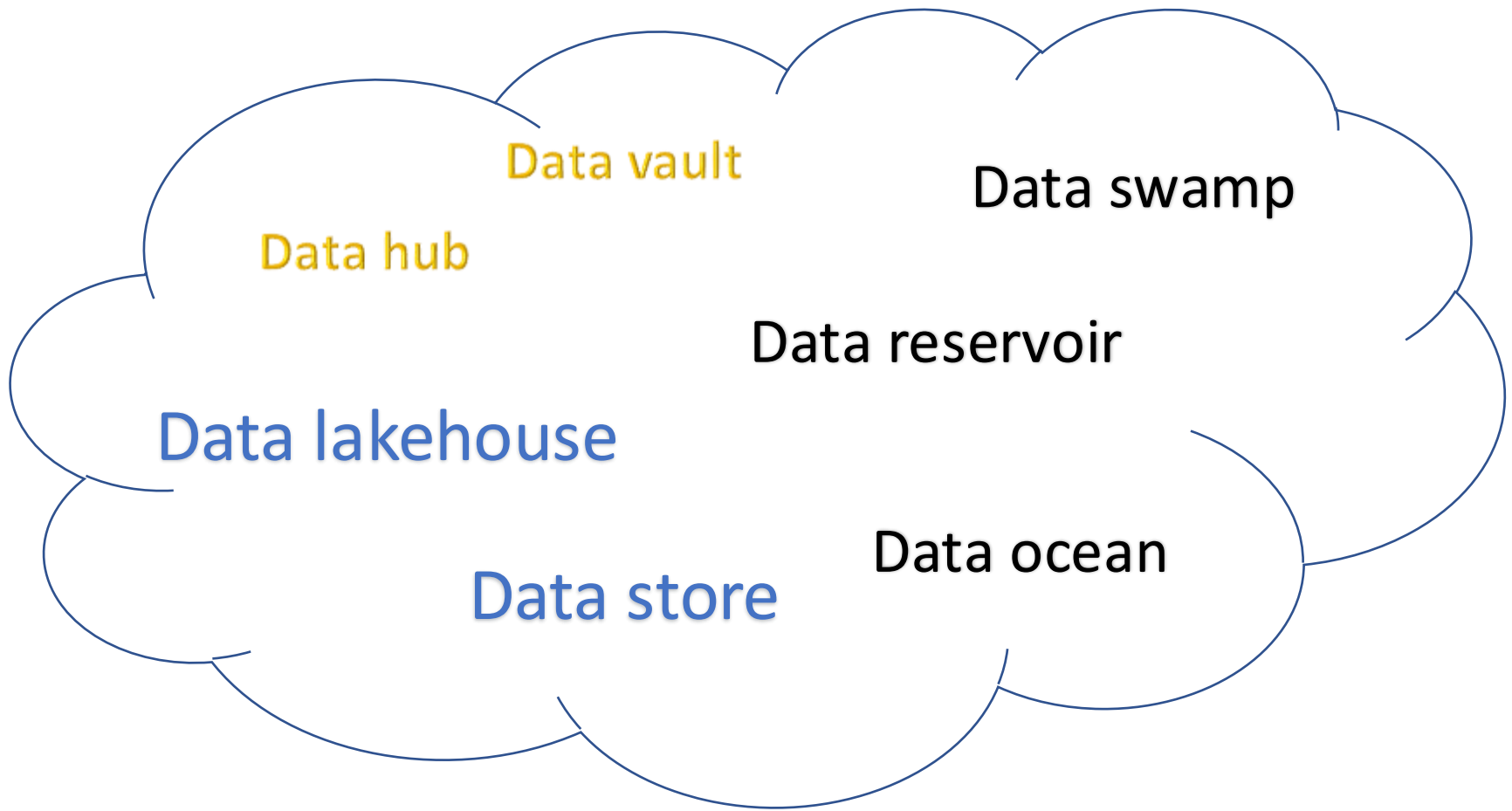


Azure
Data Lake

An example of unified framework



Have you ever seen these terms? Discover them!



ETL: Extract – Transform – Load

- **ETL** is the process of moving data from one or multiple data sources to a unified repository.
- **Extraction:** pull the raw data from different sources
 - Data could come from transactional applications or IoT sensors and in several formats (e.g., relational databases, XML, JSON).
- **Transformation:** update the data to match organizational needs and data storage solution requirements.
- **Loading:** deliver and secure the data for sharing, making business-ready data available to internal and external users

ELT: Extract – Load – Transform

- **ELT** is a variation of ETL in which data is extracted and loaded before it is transformed.
- ELT vs. ETL:



Transform
and Load

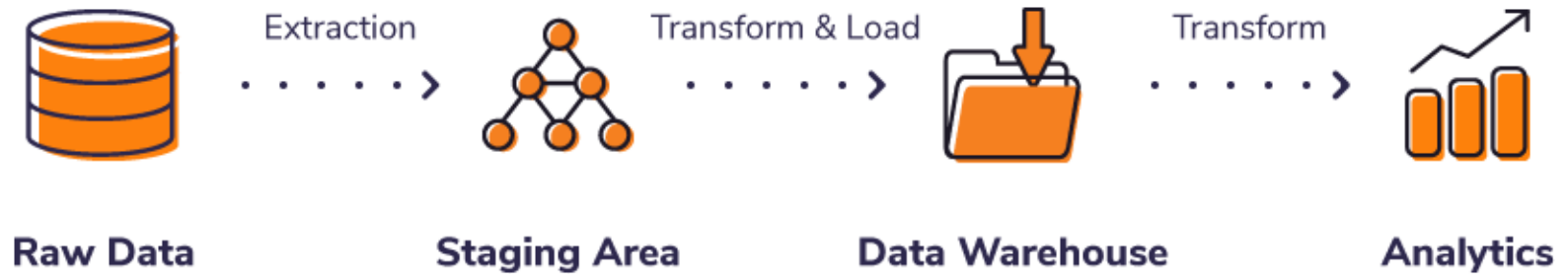


Data repository



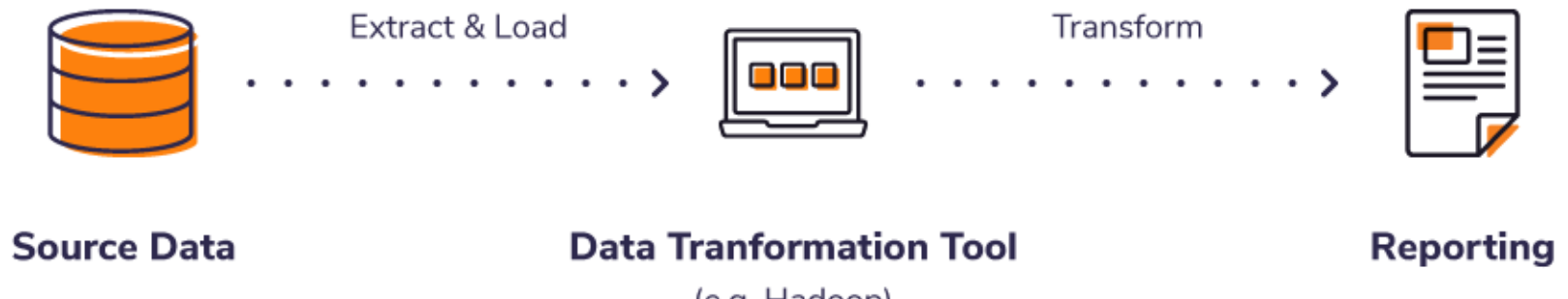
Supporting tools
and communities

The ETL Process



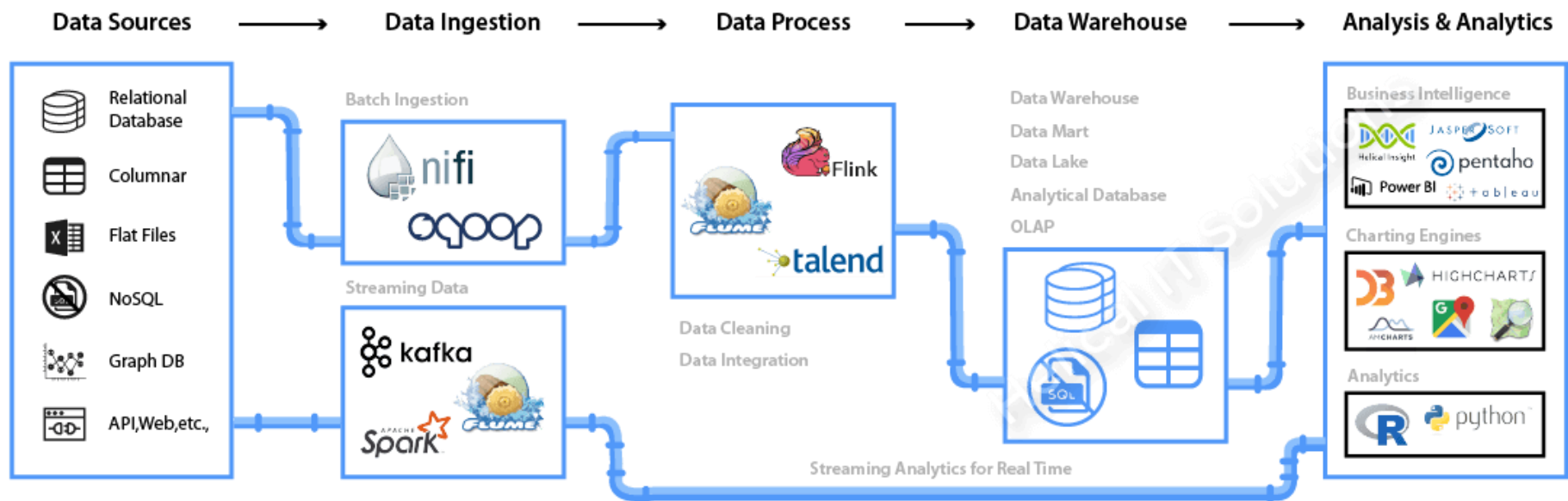
Software Advice.

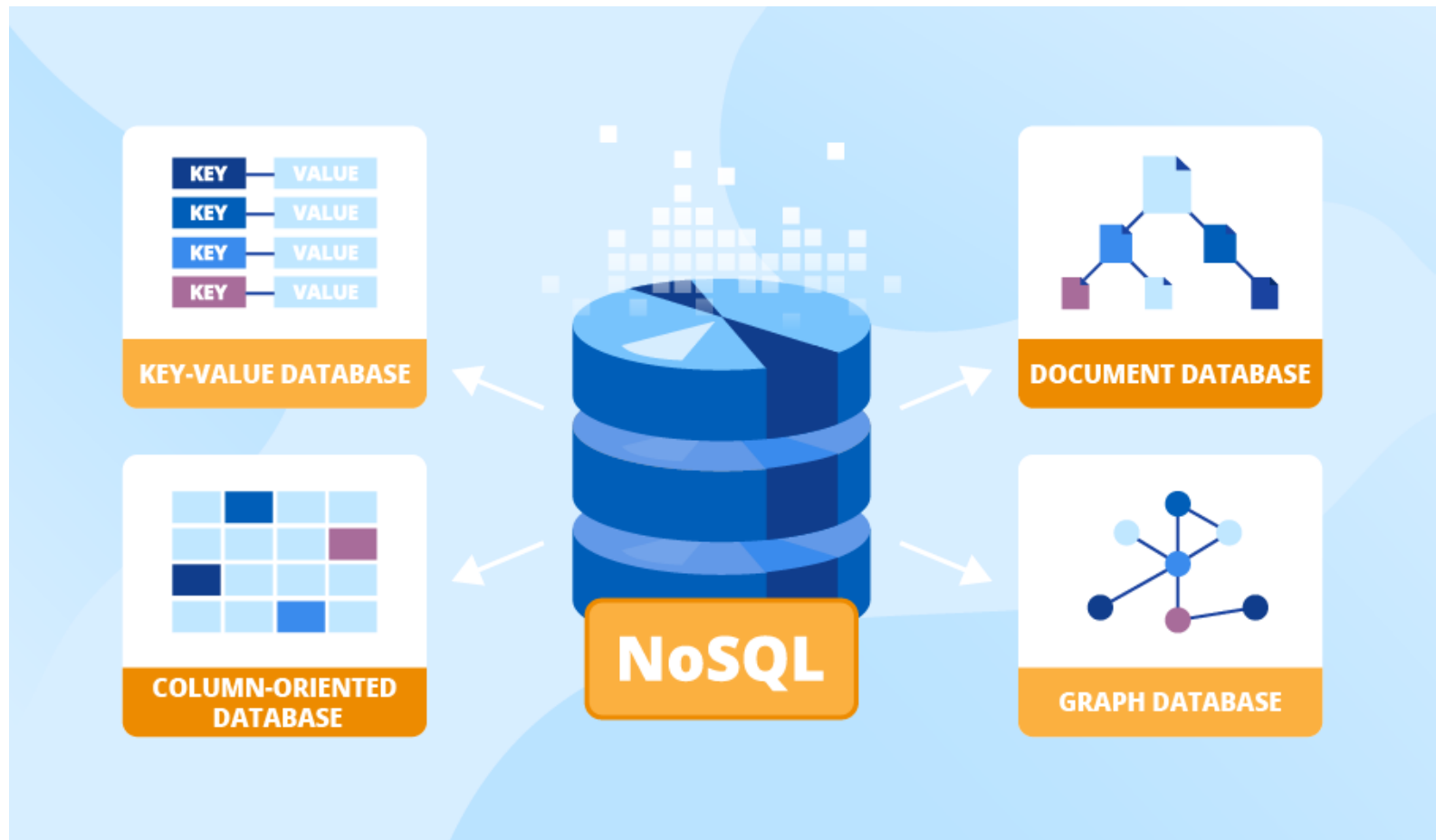
The ELT Process



Data ingestion

- **Data ingestion** is the process of driving a wide variety of data sets into where it needs to be in required format and quality.
- ETL and ELT are both viable methods, depending on your business needs

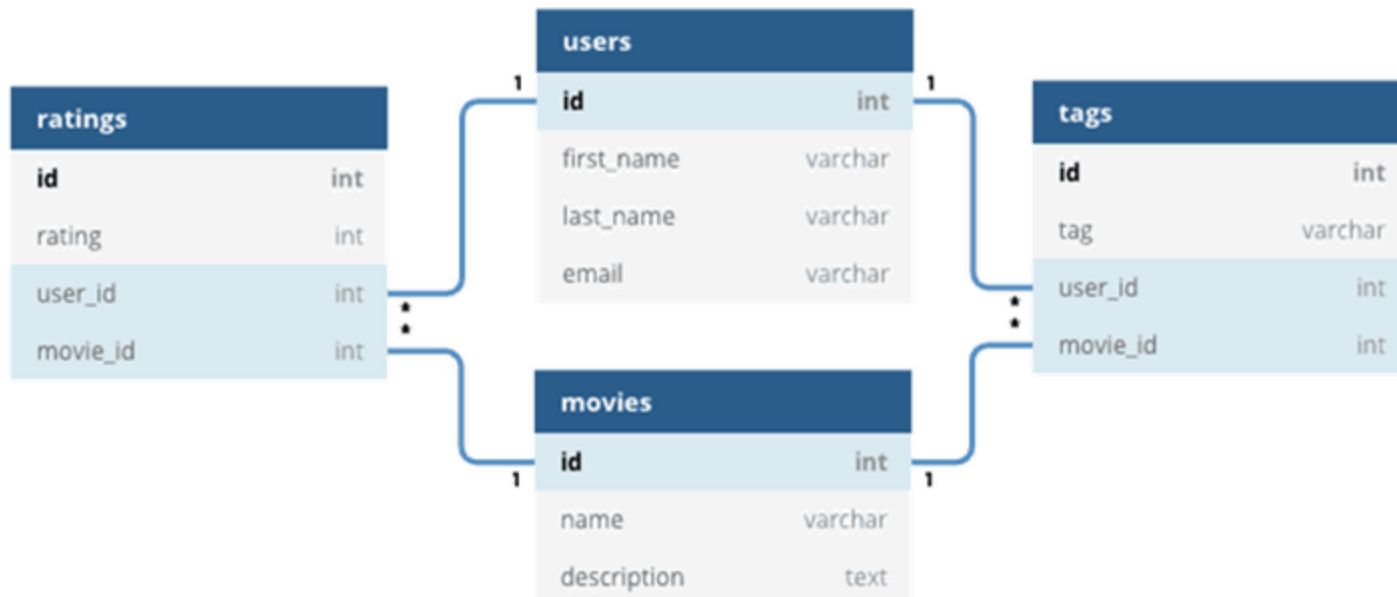




NoSQL databases

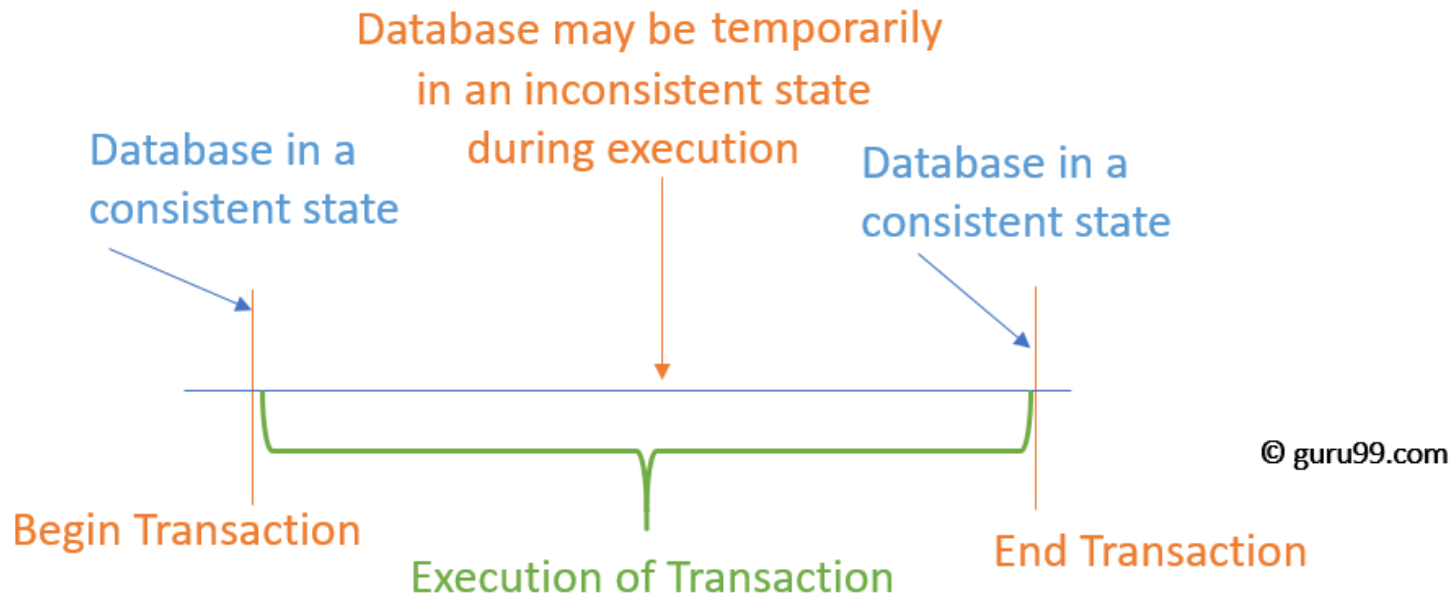
Relational databases

- **Relational DB** has been a prevalent technology for decades.
 - Mature, proven, and widely implemented
 - Competing database products, tooling, and expertise abound
- Related tables are managed in a strict schema using SQL.
- ACID guarantees are supported.



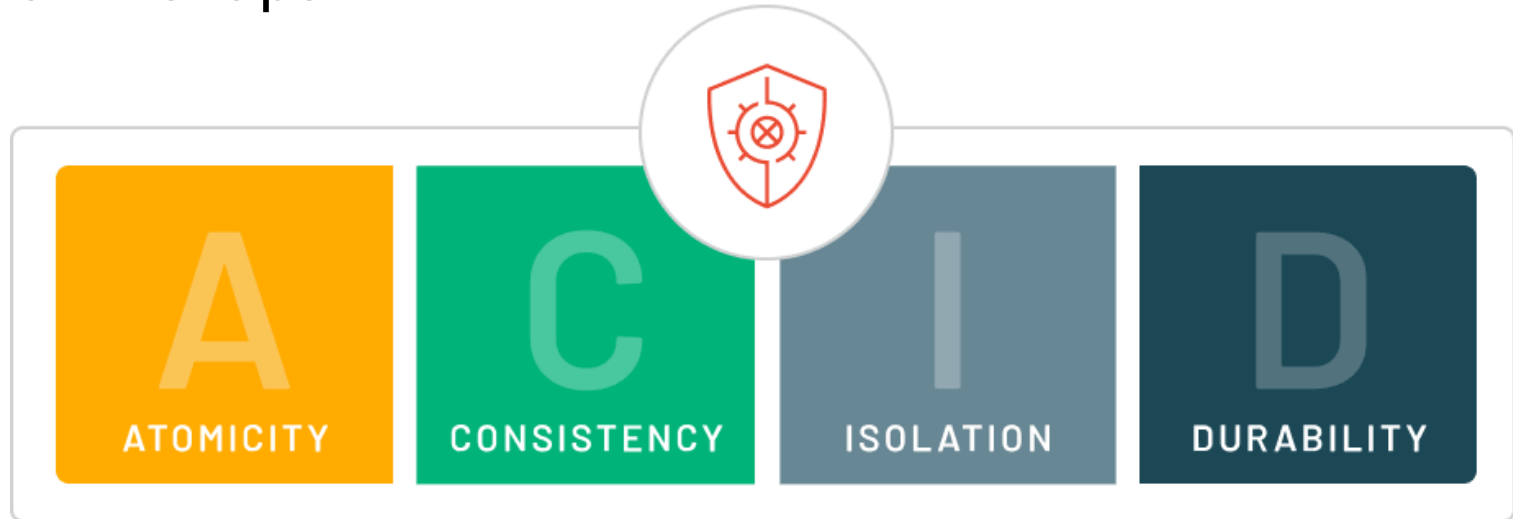
Transactions in RDBMS

- A **transaction** is any operation treated as a **single unit of work**, which **either completes fully or does not complete at all**.
 - E.g., withdraw money from bank account: either the money has left your bank account, or it has not – there cannot be an in-between state
- It leaves the storage system in a consistent state.



ACID properties

- This is a set of properties **for database transactions**.
- They **ensure data validity** despite errors, power failures, and other mishaps.



- Data storage systems that apply these operations are called **transactional systems**.

Atomicity

- A transaction is a single unit of operation. You either execute it entirely or do not execute it at all. There cannot be partial execution.
- This property prevents data loss and corruption from occurring.

Consistency

- Transactions only make changes to tables in predefined and predictable ways.
- Data errors or corruption do not create unintended effect for the integrity of tables.

Isolation

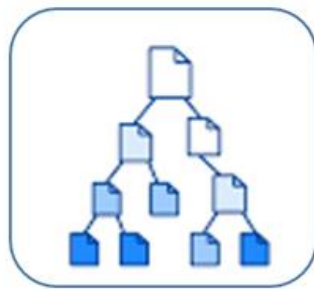
- The concurrent transactions of multiple users do not interfere with another.
- During execution, intermediate transaction results from simultaneously executed transactions should not be made available to each other.

Durability

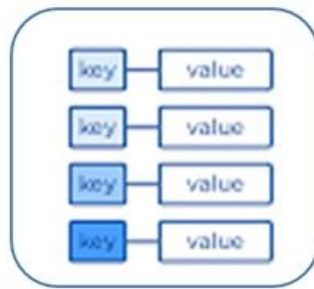
- Changes to data made by successfully executed transactions will be saved, even in the event of system failure.

What are NoSQL databases?

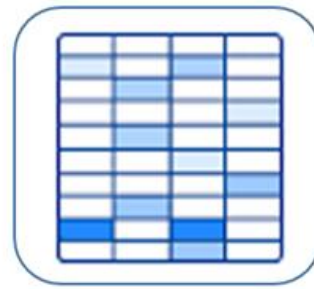
- **NoSQL databases** are **non-tabular** and they store data in a format **other than relational tables**.
- They come in a variety of types based on their data models.



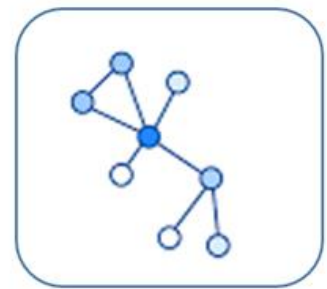
Document
Store



Key-Value
Store



Wide-Column
Store



Graph
Store

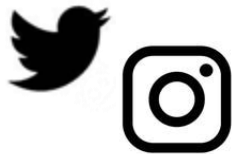
- Provide flexible schemas
- Scale easily with large amounts of data and high user loads

NoSQL databases

- NoSQL databases excel in their **ease-of-use**, **scalability**, **resilience**, and **availability characteristics**.
 - They are appropriate for high-volume services that need sub-second response time.
- They typically do **not support ACID guarantees** beyond the scope of a single database partition.
- Applications



Gaming



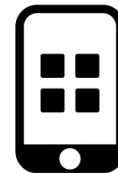
Social
network



IoT



Web



Mobile



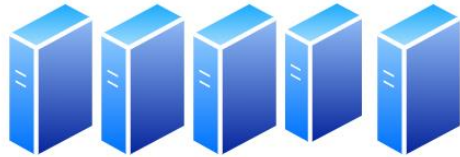
Enterprise

NoSQL databases: Key features

- Each NoSQL database has its own unique features.
- At a high level, many NoSQL DBs have the following features



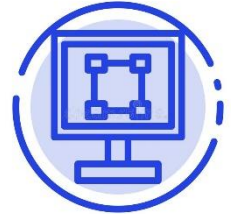
Flexible
schema



Horizontal
scaling



Fast queries due to
the data model

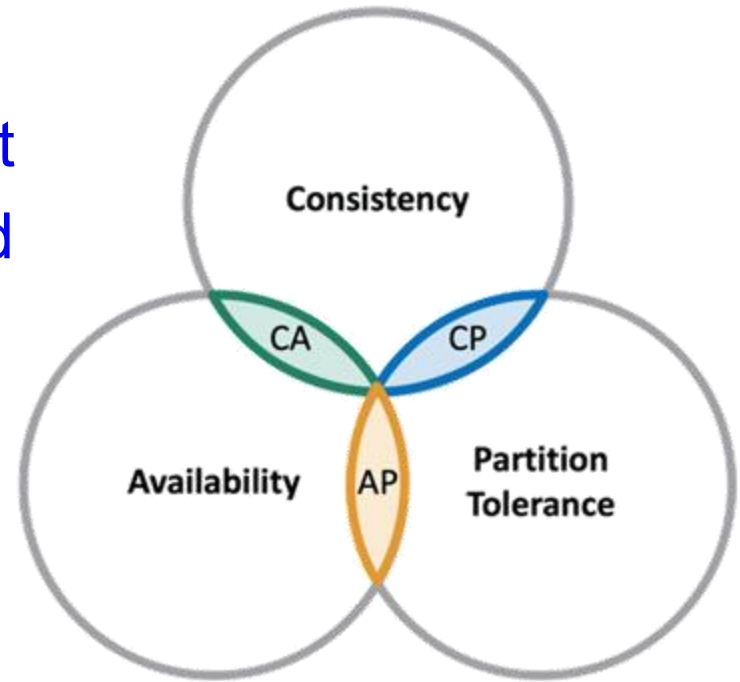


Easy development

Consider a NoSQL datastore when:	Consider a relational database when:
The workload volume is high. It requires predictable latency at large scale (e.g., latency measured in milliseconds while performing millions of transactions per second)	The workload volume generally fits within thousands of transactions per second
Data is dynamic and frequently changes	Data is highly structured and requires referential integrity
Relationships can be de-normalized data models	Relationships are expressed through table joins on normalized data models
Data retrieval is simple and expressed without table joins	You work with complex queries and reports
Data is typically replicated across geographies and requires finer control over consistency, availability, and performance	Data is typically centralized, or can be replicated regions asynchronously
Your application will be deployed to commodity hardware, e.g., with public clouds	Your application will be deployed to large, high-end hardware

The CAP theorem

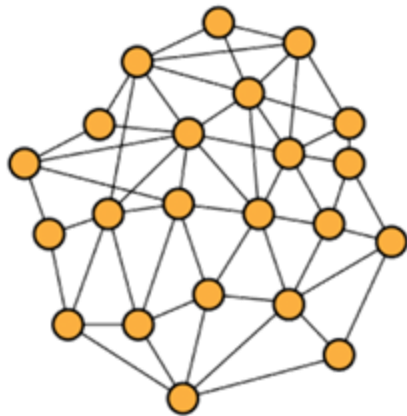
- The **CAP theorem** describes a set of principles applied to distributed systems that store state.



- It explains the tradeoffs native to managing consistency and availability during a network partition.
 - However, tradeoffs with respect to consistency and performance also exist with the absence of a network partition.

C – Consistency

At any given time, all nodes in the network have exactly the same (most recent) **value**.



● = Value: X @ 2018-05-03T08:52:40

A – Availability

Every request to the network receives a **response**, without any guarantee that returned data is the most recent.



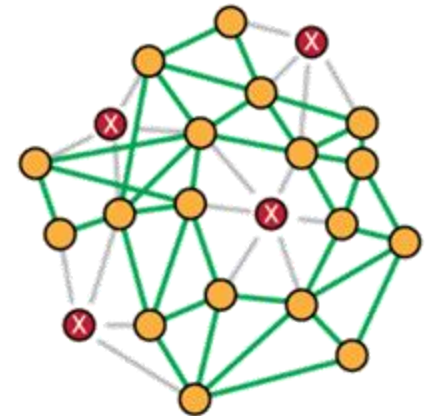
● = Value: X @ 2018-05-03T08:52:40

● = Value: Z @ 2018-05-03T08:32:58

● = Value: Y @ 2018-05-03T07:12:12

P – Partition Tolerance

The network continues to operate, even if an arbitrary number of nodes are **failing**.



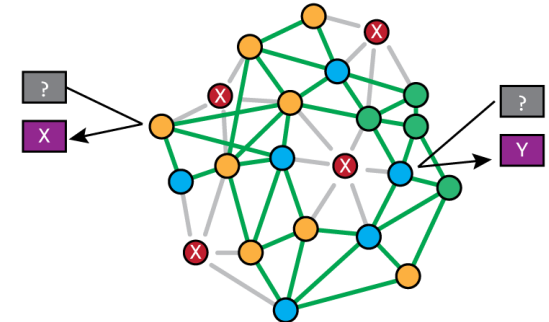
The CAP theorem

- Any database can only guarantee two of the three factors.
 - E.g., RDBMS (CA), MongoDB (CP), and Cassandra (AP)
- Relational databases are CA but not P.
 - They are typically provisioned to a single server and scale vertically by adding more resources to the machine.
- NoSQL databases are typically CP or AP.
 - Partition tolerance can never be ruled out → there will always be failing nodes in the network.
 - One must choose between C and A when in the presence of P.

Distributed systems with CAP

- Availability over Consistency (A + P)

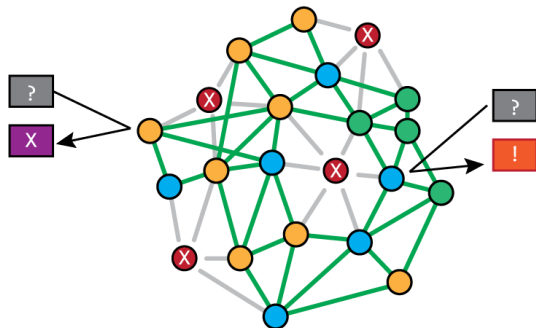
- Every request receives a response, even if the network cannot guarantee it is up to date due to network partitioning (failing nodes).
- The system will be highly available, yet not guarantee that the data returned is most recent.



● = Value: X @ 2018-05-03T08:52:40

● = Value: Z @ 2018-05-03T08:32:58

● = Value: Y @ 2018-05-03T07:12:12



● = Value: X @ 2018-05-03T08:52:40

● = Value: Z @ 2018-05-03T08:32:58

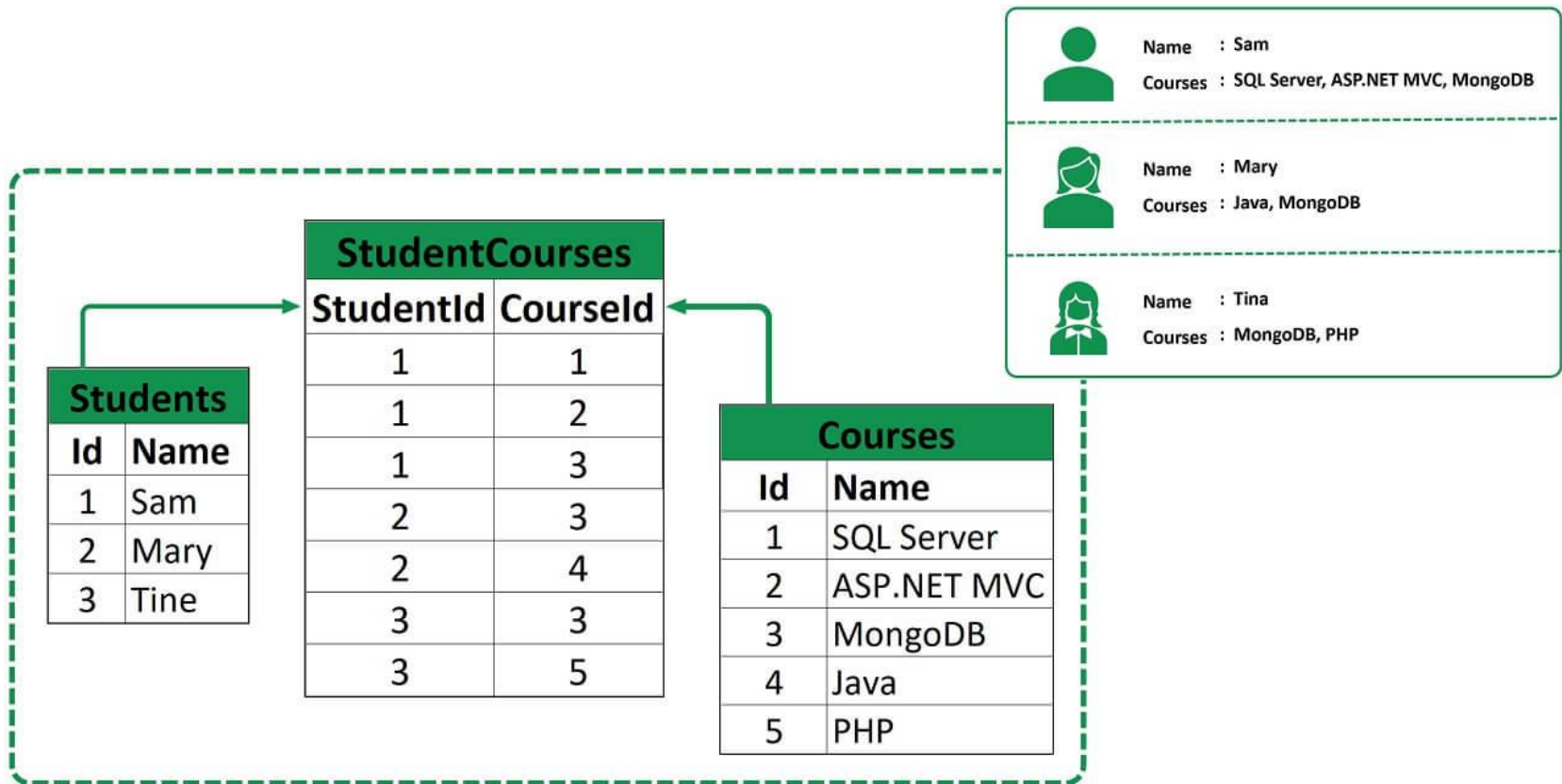
● = Value: Y @ 2018-05-03T07:12:12

- Consistency over Availability (C + P)

- The system will return an error or a time-out if information cannot be guaranteed to be up to date due to network partitioning.
- The system will be highly accurate, yet most likely be unavailable for 99.99% of the time.

NoSQL databases: Misconceptions

- NoSQL databases do not store relationship data well.
- They just store relationship data differently than RDBMS do.

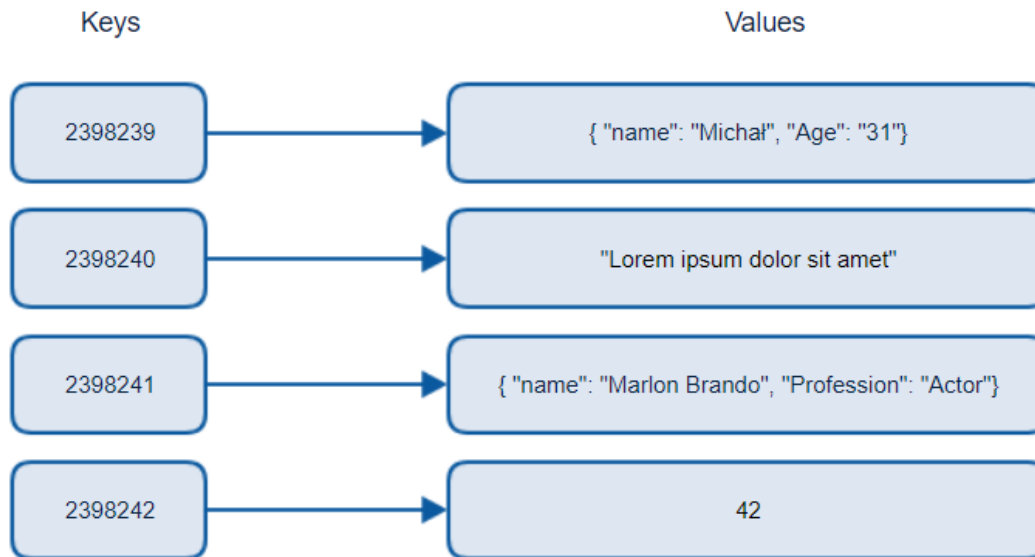


NoSQL databases: Misconceptions

- NoSQL databases do not support ACID transactions.
- Many NoSQL databases do, maybe not fully support.
- The way data is modeled can eliminate the need for multi-record transactions in many use cases.

Data models: Key-value store

- The simplest type of NoSQL databases
- The data is represented as **a collection of key-value pairs.**



Keys and values in Azure Cosmos DB



Data models: Document store

- These databases store the data and metadata **hierarchically** in **JSON-like documents**.

RELATIONAL

Peer_ID	Surname	FirstName	City
0	Sharma	Ram	London
1	Gupta	Shyam	Delhi
2	Chauhan	Bobby	Kolkata
3	Dalal	Mukesh	Gurgaon

PERSON

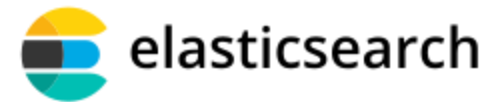
Car_ID	Model	Year	Value	Peer_ID
101	Bently	1990	8000000	0
102	Caddilac	1995	3000000	0
103	Renault	2008	800	1
104	Smart	1999	2000	2

Cars



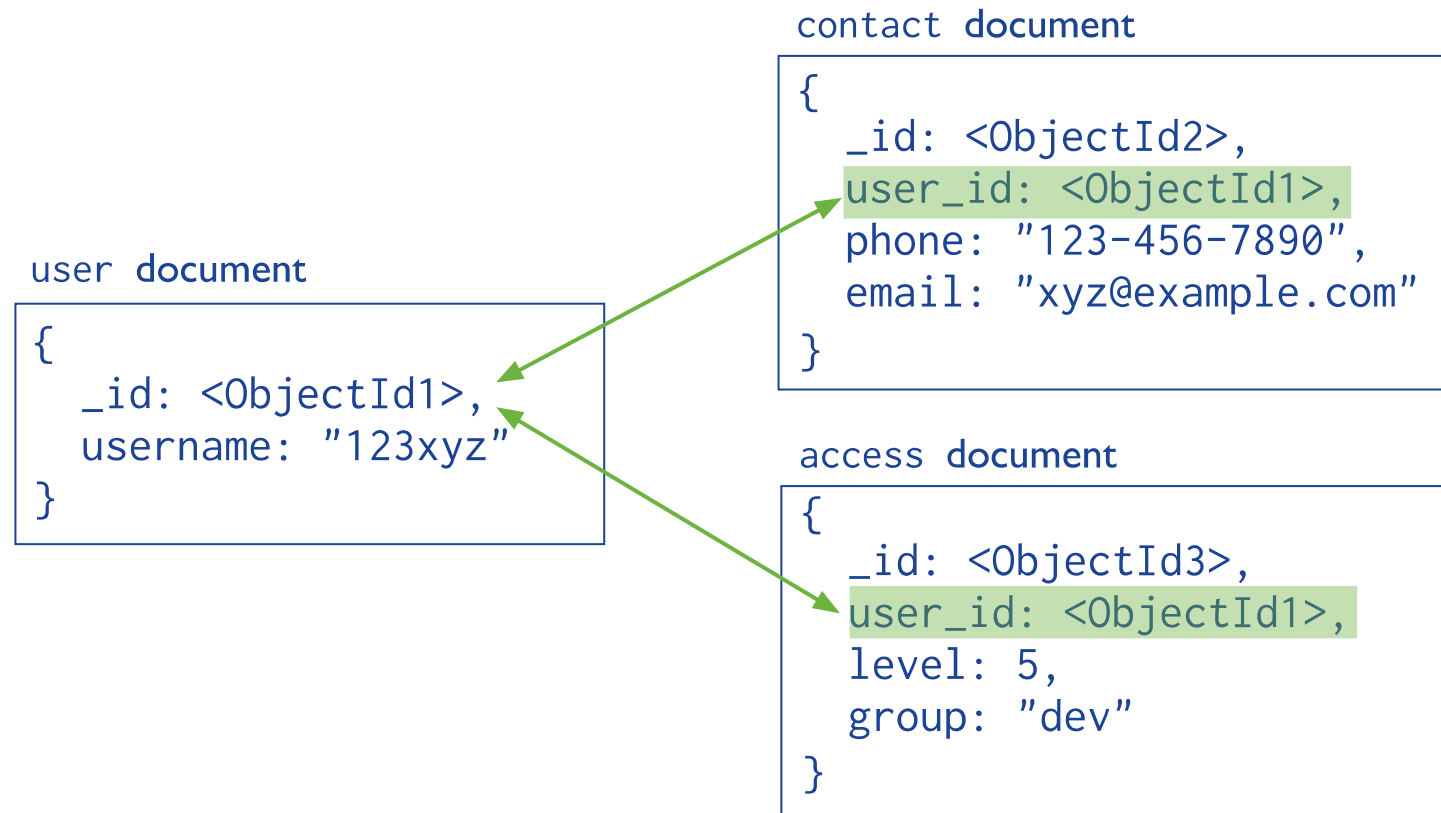
MongoDB Document

```
{
  FirstName : 'Ram',
  Surname : 'Sharma',
  City : 'London',
  Cars : [
    { Model : 'Bently',
      Year : 1990,
      Value : 8000000
    },
    { Model : 'Caddilac',
      Year : 1995,
      Value : 3000000
    }
  ]
}
```



Normalized data models

- Relationships are depicted by **references between documents**.



Denormalized data models

- Related data is **embedded in a single structure** or document.
- This involves simple operations to manipulate related data.



Data models: Wide-column store

- The data is stored as a set of nested-key/value pairs within a single column.

Row A	Column 1	Column 2	Column 3	...
	Value	Value	Value	

Row B	Column 2	Column 3	Column 4	...
	Value	Value	Value	



Data models: Graph store

- We use graph structures (e.g., node, edge, and properties) to describe the data.



Multi-model databases

- **Multi-model databases** are designed to support **multiple data models** against a single, integrated backend.

	Key-value	Document	Graph	Streaming
 Franz Inc. AllegroGraph	☐	☒	☒	☐
 ArangoDB	☒	☒	☒	☐
 redis	☒	☒	☒	☒

In-memory databases

- **In-memory databases** primarily store data on main memory, which contrasts to those using physical disks.
- Minimal response time is enabled by eliminating the need to access disks.





Amazon Elasticsearch Service



Fully Managed



Petabyte Scale



Secure



Highly Available

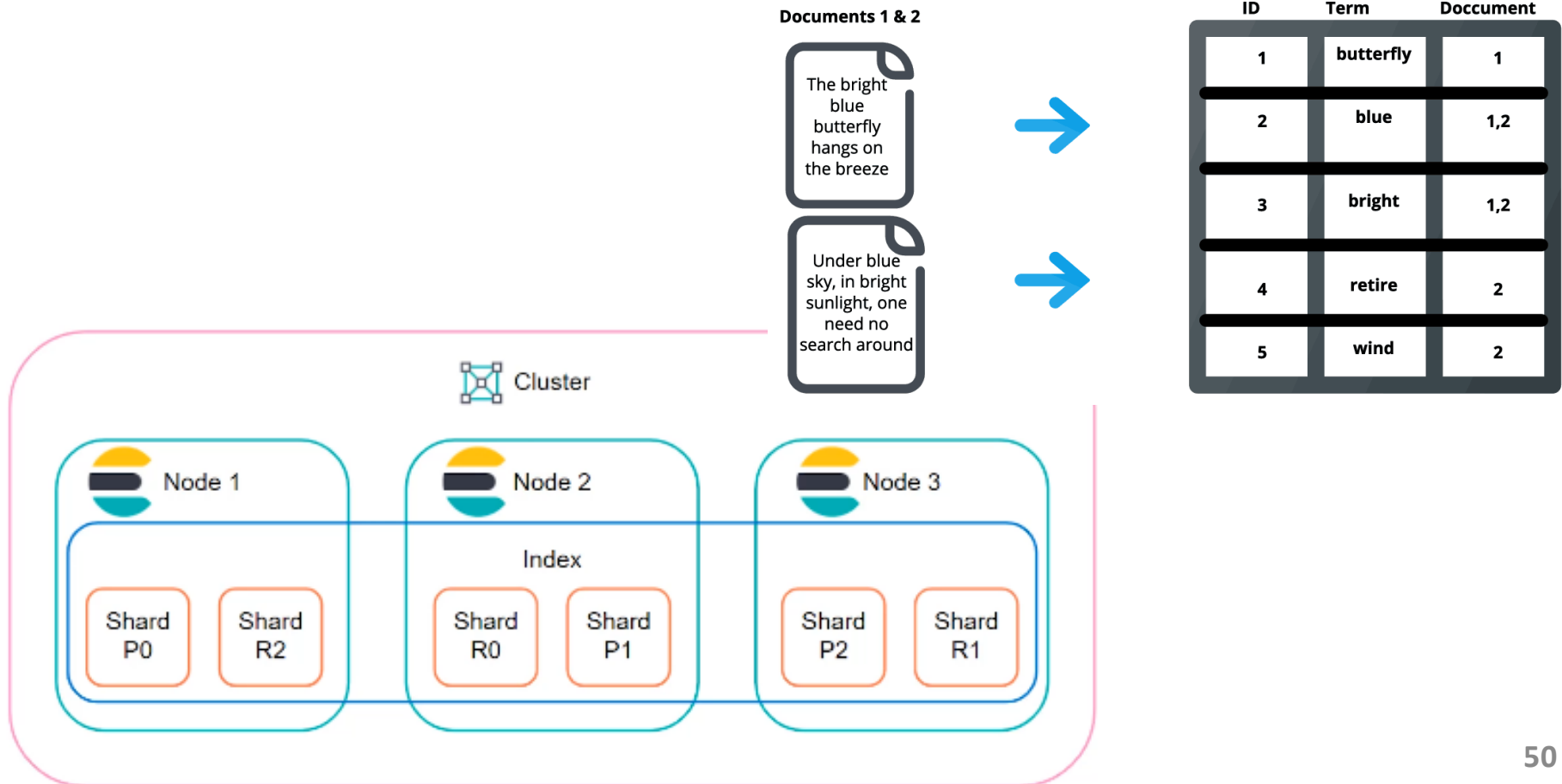


Scalable

A distributed, multitenant-capable full-text search engine with an HTTP web interface and schema-free JSON documents

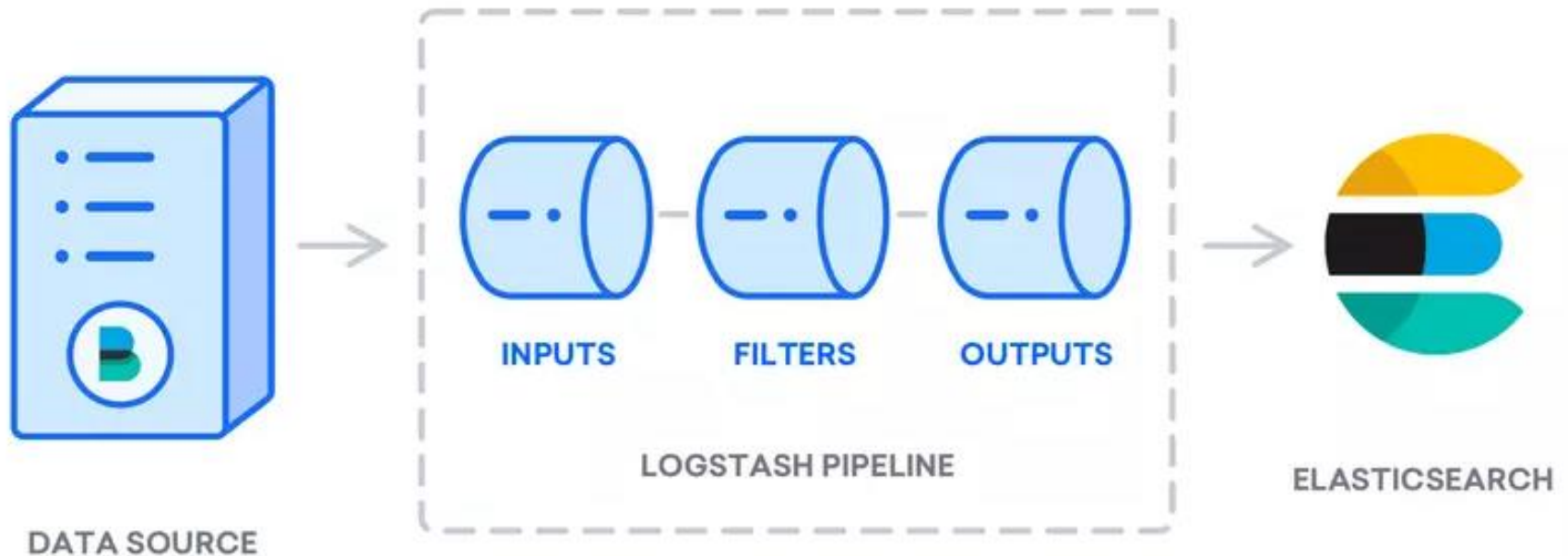
Elasticsearch cluster

- The power of an Elasticsearch cluster lies in the **distribution of tasks, searching, and indexing**, across all the nodes.



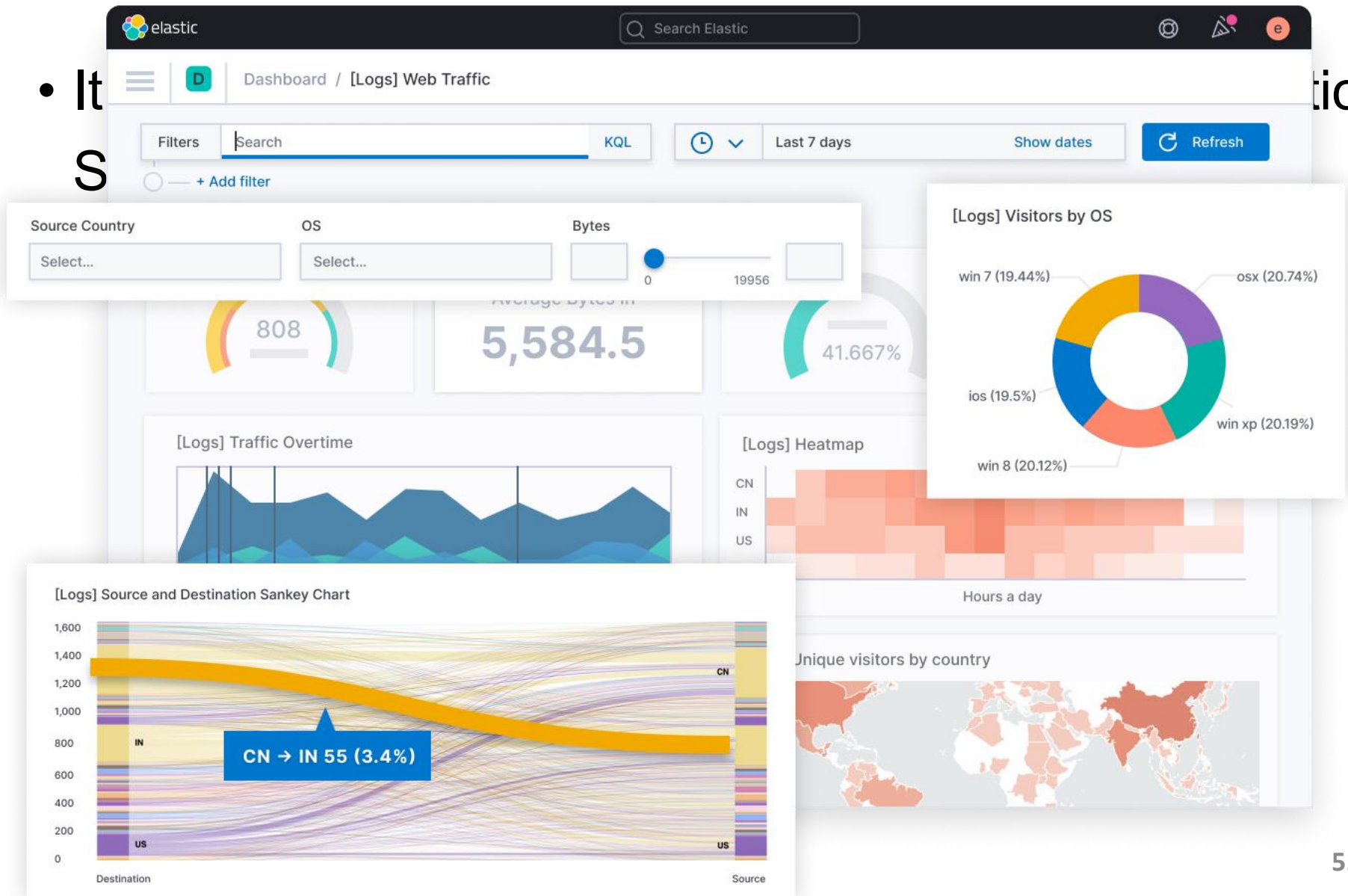
Elasticsearch: Logstash

- **Logstash** ingests data from various sources, transforms it, and sends it to Elasticsearch.



Elasticsearch: Kibana

- It's



...the end.

